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Not much "I" in "Team": Content analysis of pregame speeches in American football

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Abstract

Although researchers have explored the effects of coach pregame speeches, little is known about the actual content, delivery, and environmental context of the speeches themselves. This study was a descriptive analysis of 127 ($n = 77$ intercollegiate, $n = 37$ high school, and $n = 13$ cinematic) American football pregame speeches. The speeches were viewed, transcribed, coded, and analyzed via the Linguistic Inquiry and Word Count (LIWC2015) software program. Actual speeches were compared with natural speech norms and speeches appearing in movies pertaining to American football. In general, the pregame speeches were characterized by use of confident, analytical, motivational, emotionally tinged language that is collectively oriented and present and future focused. The speeches delivered to high school teams were substantially similar to those delivered to intercollegiate teams. The actual speeches differed from natural speech on numerous variables and from the cinematic speeches on some descriptive variables. The findings suggest that actual American football pregame speeches are fundamentally similar to cinematic American football pregame speeches and constitute a distinct form of communication that is tailored to the unique demands of the situation.

Keywords

Coaches, communication, emotion, inspiration, motivation, psycholinguistic, team sport

In competitive sport, there is perhaps no more quintessential application of psychology than the pregame speech. Common practice in the team sport environment,[1,2] pregame speeches involve coaches (or other team affiliates) delivering a final verbal message to a team prior to the beginning of a competition. It has been proposed that pregame speeches may enhance mental preparedness of athletes for competition by providing encouragement, furnishing technical and tactical information, boosting confidence/self-efficacy, increasing motivation, and inducing facilitative emotions and physiological arousal.[1–6]

Despite the ubiquity and potential practical importance of pregame speeches, they have received relatively little attention in the empirical literature. In a pair of studies of the effects of simulated coach pregame speeches, athletes rated speeches with emotional content as producing greater team efficacy,[7] inspiration to perform, and emotional dominance[1] than those with informational (i.e. instructional and strategic) content. Athletes' preferences for emotional and informational content in pregame speeches appear to vary across competitive situations, as demonstrated in an investigation of hypothetical pregame speeches.[6]

Studies involving actual, rather than simulated, pregame speeches have also been conducted. These studies have focused on the effects of pregame speeches on factors such as self-efficacy rather than on the content of the speeches themselves. Increases in self-efficacy have not typically been documented in investigations featuring pretest–posttest designs to assess changes in variables from before pregame speeches to after pregame speeches.[2,5] In one study, however, increases in self-efficacy and functional emotions were positively associated with athletes' perceptions of the amount of informational content in their coach's pregame speech.[2] In studies where athletes' perceptions of their coaches' pregame speeches have been

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examined, athletes have expressed general satisfaction with the speeches,[4,8] reported being positively affected by the speeches in terms of increased effort and facilitative psychological states, and to contain informational and emotional content.[8] Perceived characteristics of speeches described as favorable were brevity, meaningfulness, expression of genuine emotion, and consistency with team values, whereas speeches perceived as long, poorly timed, and lacking in information were described unfavorably.[4]

Aside from a study in which Turman[9] examined the types of regret messages used by coaches in pregame speeches (and other communication with athletes), there has been minimal investigation of the actual (rather than perceived) content of pregame speeches. Consequently, little is known about even the most basic aspects of pregame speeches, such as their content, their delivery, and the environmental context in which they are given. In light of these gaps in the literature, the primary purpose of the current study was purely descriptive in nature, involving a detailed analysis of the content, delivery, and environmental context of pregame speeches. To reduce the potential influence of sport-related factors on the content of pregame speeches, we focused on speeches from a single sport: American football. Characteristics of the speakers, the behavior of speakers during the pregame speeches, the setting in which the pregame speeches were given, and the pregame speeches themselves (e.g. duration, location, topical references, and rhetorical devices) were assessed in an attempt to describe the phenomenon of interest. Given the preliminary and descriptive nature of the study, no formal hypotheses were made. It was, however, anticipated that the speeches would: (a) feature both informational and emotional content[1,2,6–8]; (b) correspond to the arousing, confidence-boosting, and motivating functions ascribed to pregame speeches[1–6]; and (c) reflect a situational context in which a (presumably) credible speaker is addressing an intact group (i.e. a team) prior to a game with a clearly defined opponent.

A secondary purpose of the study was to examine several comparative questions that could be addressed using data collected during the course of the descriptive analysis. Specifically, whether and how pregame speeches might be tailored to different athlete populations and might differ from both ordinary speech and popularized renditions of pregame speeches as portrayed in motion pictures were explored. The issue of potential tailoring of pregame speeches to the audience was examined by comparing the speeches given to high school teams with those given to intercollegiate teams, thus allowing an assessment of similarities and differences of speeches addressed to players in the same sport but of different ages (and competitive levels). Comparing actual speeches with ordinary speech and pregame speeches featured in motion pictures facilitated an assessment of the distinctiveness of pregame speeches as a form of discourse and the extent to which real-world pregame speeches may shape or have been shaped by cinematic versions of pregame speeches, respectively. Such comparisons have the potential to shed light on what pregame speeches are and are not.

Method

Data collection

Video clips depicting pregame speeches delivered in movies featuring American football and to high school and intercollegiate American football teams were identified on the YouTube online video platform (www.youtube.com) with the search term "football pregame speech" on January 1, 2018. YouTube videos that included complete, audible, and visually available American football pregame speeches were downloaded and transcribed for further analysis. A total of 127 ($n = 77$ high school, $n = 37$ intercollegiate, and $n = 13$ cinematic) speeches met the inclusion criteria. Twenty-three videos were excluded due to being incomplete ($n = 15$), inaudible ($n = 6$), or visually unclear ($n = 2$).

Although it is not known how representative the pregame speeches posted to YouTube are of those across the full population of pregame speeches delivered to high school and intercollegiate football teams, YouTube was used as the source of data because it yielded a large and diverse sample of pregame speeches that were publicly available, readily accessible, and not subject to the geographical and logistical constraints typical of offline data sources. Consequently, the YouTube data were deemed an appropriate point of departure for descriptive analyses of pregame speeches.

Data coding

Two coding systems were applied to the data collected from YouTube, one designed specifically to describe aspects of pregame speeches (i.e. descriptive and verbal content coding system) and one designed to describe speech in general (i.e. Linguistic Inquiry and Word Count 2015). Although there was partial overlap between the two systems, they both provided unique insights that, when combined, facilitated a comprehensive analysis of the content and delivery of pregame speeches.

Descriptive and verbal content coding system. A video coding system was developed to allow for systematic coding of descriptive and verbal content of the pregame speeches. The coding system was developed through a bottom-up process in which: (a) a sample of pregame speeches was viewed; (b) salient, codable aspects of the content, delivery, and environmental context of the pregame speeches were identified; and (c) categories, codes, and coding instructions were refined over multiple iterations. Development of the

coding system was led by the first author, who (a) gathered the sample of pregame speeches for the second author, third author, and other members of the research team to view; (b) completed preliminary drafts of the categories, codes, and coding instructions; and (c) incorporated feedback from the second author, third author, and other members of the research team into the final version of the system that was used in the study. As displayed in Table 1, 16 descriptive and verbal content variables were coded for each speech. Coders were trained to criterion with respect to knowledge of variable terms and coding ability. After reviewing the pregame speech analysis manual developed for the current study, coders were required to demonstrate their competence by completing a 17-item assessment on variable terms and raw-data input into a coding data sheet. Coders then viewed and coded 5 cinematic pregame speeches. Coders who met the standards of the manual coded the study data.

Linguistic Inquiry and Word Count 2015. The Linguistic Inquiry and Word Count (LIWC)[10] software application was developed by Francis and Pennebaker[11] to facilitate empirical investigation of language and discourse. The software has been updated and expanded over the past three decades[12–14] and has been widely used across multiple disciplines and topic areas.[15–19] The 2015 version of the software (i.e. LIWC2015) was used in the current study to analyze the verbal content of pregame speeches along multiple linguistic, grammatical, and psychological dimensions. LIWC2015 analyses yield four summary variables that are reported as standardized scores ranging from 0 (very low) to 100 (very high): analytical thinking, clout, authentic, and emotional tone. Analytical thinking reflects formal, logical, and hierarchical thinking versus informal, personal, here-and-now, and narrative thinking. Clout indicates whether the speaker is speaking from a perspective of high expertise and confidence or is more tentative, humble, or anxious. Authentic assesses the extent to which the speaker uses language that is honest and personal. Emotional tone indicates whether a speaker is positive and upbeat or anxious, sad, or hostile.[20] The summary variables were used to assess and describe the language used in pregame speeches, and to help determine the extent to which the language used in pregame speeches exuded the confidence that pregame speeches are often intended to instill (clout) and correspond with the strategic (analytical thinking) and emotional (emotional tone) functions ascribed to pregame speeches.

From among the more than 90 additional variables assessed by LIWC2015, 13 were deemed of potential relevance to pregame speeches and examined in the current study: compare, positive emotion, negative emotion, drives, past focus, present focus, future focus, swear words, and 5 personal pronouns (i.e. "I," "We," "You," "He/She," and "They") assess distinctions among individuals or groups in a pregame speech. Compare refers to words the speaker uses to compare one entity (e.g. team, player, and season) with another. This variable was assessed in light of the potential for comparisons with the team's opponent in pregame speeches. Positive emotion is the speaker's choice of words/phrases that elicit the experience of positive emotions, whereas negative emotion is the

Table 1. Descriptive and verbal content variables.

Variable	Description
Speech length	Duration of pregame speech
Speaker	The individual performing the pregame speech
Speaker-player contact	Speaker makes physical contact with member(s) of the team at least once
Speech location	Setting of the pregame speech
Speaker location	Position of the speaker, relative to the team, when delivering the speech
Opponent reference	Words and/or phrases that mention/refer to the team against whom the listeners of the pregame speech would be competing
Repeated word/ phrase	A word or phrase that is said multiple times with emphasis by the speaker during the pregame speech
In-speech instruction	An instruction given by speakers to the team to be carried out during the pregame speech
Game instruction	An instruction pertaining to performance and/or strategy for the upcoming competition
Questioning	A rhetorical or responsive question posed by the speaker to the team and/or one or more specific players
Team-past reference	Phrases or allusions that refer to the team in the time prior to the pregame speech
Future reference	Phrases or allusions that refer to the team in the time after the game about to be played
Environment reference	Words, phrases, or allusions that refer to the field, stadium, crowd, community, etc.
Metaphor	A figure of speech in which a word or phrase literally denoting one kind of object or idea is used in place of another to suggest likeness or analogy between them
Anecdote	Narrative of a personal past event used by the speaker

speaker's choice of words/phrases that elicit the experience of negative emotions. The positive and negative emotion variables were included to gain a more fine-grained analysis of the emotional content of pregame speeches than that provided by the emotional tone summary score. Drives language pertains to references to motivational content (e.g. wins, goals, and rewards) by the speaker, potentially corresponding to the motivational function of pregame speeches. Past focus, present focus, and future focus assess verbs and references to events/times corresponding to the past, present, and future, respectively. The pregame aspect of pregame speeches made future focus a variable of potential relevance. Swear words are profane, obscene, and/or offensive word(s) used as an expression of anger or frustration. This variable was considered potentially relevant due to the age discrepancy in the audiences for high school and intercollegiate pregame speeches. Personal pronouns were considered useful for assessing the extent to which the language used in pregame speeches reflected the collective endeavor at hand (i.e. the team's involvement in a game). Scores for each category variable are given as a percentage based upon the number of words for each variable compared to the entire pregame speech transcript.

Pennebaker et al.[10] assessed the internal reliability of LIWC2015 by measuring the use of variable category words included in the software's dictionaries across multiple text files. Pennebaker and Francis[17] validated the scales of the LIWC by comparing judges' ratings of emotional, cognitive, and content dimensions with the LIWC output of college students' essays on various topics. The findings of the study suggested that the LIWC "successfully measured positive and negative emotions, a number of cognitive strategies, several types of thematic content, and various language composition elements" (Pennebaker et al.,[10] p. 8). LIWC has been used successfully in hundreds of studies.[21]

Data analysis

Transcriptions of the speeches were uploaded to LIWC2015 for analysis. LIWC2015 used its database of dictionaries for specific linguistic variables and summary variable algorithms to analyze the linguistic content of each pregame speech transcript. Data from the LIWC2015 and the coding of descriptive and verbal content were inputted into SPSS for statistical analysis. Frequencies were calculated for the descriptive and verbal content variables, and means and standard deviations were computed for the LIWC2015 variables. Chi-square analyses were conducted to compare the actual pregame speeches with the cinematic pregame speeches and the intercollegiate pregame speeches with the high school speeches on the descriptive and verbal content variables. The same comparisons were made for the LIWC2015 variables with two pair of one-way multivariate analyses of variance (MANOVAs), with one of each pair examining the four summary scores (i.e. analytical thinking, clout, authentic, and emotional tone) and the other examining the additional 13 LIWC2015 variables of potential relevance to pregame speeches. The actual pregame speech LIWC2015 variable mean scores were also compared to "natural speech" LIWC2015 mean scores in a series of one-sample t-tests. Significance levels for the t-tests were .0125 (i.e. .05/4) for the four LIWC2015 summary variables and .0038 (.05/13) for the 13 additional LIWC2015 variables of potential relevance to pregame speeches.

Results

Descriptive and verbal content variables

The mean length of the actual pregame speeches was 123.03 (standard deviation (SD) = 66.23) s. High school pregame speeches ($M = 130.92$, $SD = 61.68$ s) did not differ significantly in length from intercollegiate pregame speeches ($M = 106.59$, $SD = 72.98$ s, $t(112)$ 1.85, $p = .07$, $d = 0.37$. The mean length of the cinematic speeches was 91.85 ($SD = 53.39$) s. The actual pregame speeches were significantly longer than the cinematic pregame speeches, $t(138) = 2.24$, $p = .03$, $d = 0.47$. The speaker in the actual pregame speeches was a coach in 83 (73%) of the speeches, a player in 11 (9%) of the speeches, and another person (e.g. military veteran, former coach, alumnus) in 14 (12%) of the speeches. It was not possible to determine who the speaker was for 6 (5%) of the actual speeches. In the cinematic speeches, the speaker was a coach in 11 (85%) of the speeches and another person in 2 (15%) of the speeches. In the actual pregame speeches, 16 (14%) featured physical contact (e.g. hand-on-shoulder-pads, hand-to-hand) between the speaker and players of the team. Speaker-player contact was evident in 3 (23%) of the cinematic pregame speeches. Actual speeches were given in a locker room for 90 (79%) of the speeches, a meeting room for 13 (11%) of the speeches, on the field for 5 (4%) of the speeches, and some other location for 5 (4%) of the speeches. Speakers in the actual pregame speeches were positioned in front of the team in 62 (54%) of the speeches, in the middle of the team in 45 (39%) of the speeches, and in another location (e.g. walking around) in 7 (6%) of the speeches. Speakers in the cinematic pregame speeches were positioned in front of the team in 9 (69%) of the speeches and in the middle of the team in 4 (31%) of the speeches.

As shown in Table 2, verbal content characteristics present in more than half of the actual speeches were opponent reference, team past reference, environment reference, repeated words, game instructions, and questioning. All of the remaining characteristics (i.e. anecdote, future reference, in-speech instructions, and metaphor) were present in at least one-quarter of the actual speeches. For the cinematic speeches, verbal content characteristics

Table 2. Frequencies of descriptive and verbal content variables for pregame speeches.

		Actual			Cinematic		Intercollegiate			High school	
Variable	Example	Yes	No	χ^{2a}	Yes	No	Yes	No	χ^{2b}	Yes	No
Group talk	"One, two, three…." [said in unison]	46	68	0.87	6	7	13	24	0.62	33	44
Opponent reference	"They're bigger, they got athletes, they got guys going to Division I schools."	69	45	5.76*	5	8	21	16	0.33	48	29
Team-past reference	"You guys had a great week of preparation."	72	42	11.31**	3	10	24	13	0.07	48	29
Anecdote	"Just remembered. When I played, I feared the bird."	32	82	1.57	1	12	10	27	0.04	22	55
Future reference	"We want to practice next week."	31	83	0.08	4	9	7	30	1.89	24	53
Environment reference	"This is our house."	78	36	0.10	9	4	23	14	0.18	51	26
Repeated words	"See a lot, see little. See a lot, see nothing."	114	0	—-	13	0	37	0	—-	77	0
In-speech instructions	"Everybody take a knee."	47	67	0.76	7	6	13	24	0.84	34	43
Game instructions	"Control the line of scrimmage."	107	7	4.61*	10	3	34	3	0.37	74	4
Questioning	"Do you believe?"	76	38	0.56	10	3	18	19	8.00**	58	19
Metaphor	"We put together an orchestra."	53	61	9.80**	12	1	18	19	0.10	35	42

*$p < .05$, **$p < .005$.
Note: χ^{2a} refers to the comparison between actual and cinematic speeches and χ^{2b} refers to the comparison between intercollegiate and high school speeches.

Table 3. Means and standard deviations of LIWC2015 variables for actual pregame speeches, cinematic pregame speeches, and natural speech.

	Actual		Cinematic			Natural
Variable	*M*	*SD*	*M*	*SD*	η_p^2	*M*
Analytical thinking	35.31	17.90	30.30	21.80	.01	18.43**
Clout	94.25	11.92	94.25	4.46	.00	56.27**
Authentic	55.53	21.62	53.03	24.39	.00	61.32*
Emotional tone	71.39	27.26	58.72	32.40	.02	79.29**
I	2.70	2.73	3.36	1.37	.01	7.03**
We	4.04	2.46	4.30	1.85	.00	0.87**
You	5.85	2.57	5.89	3.15	.00	4.04**
He/She	0.40	0.70	0.41	0.93	.00	0.77**
They	1.17	1.26	0.76	1.94	.00	0.65**
Compare	1.91	1.28	1.09	1.19	.03	2.35**
Positive emotion	4.62	2.12	3.54	2.11	.03	5.31**
Negative emotion	1.65	1.59	1.64	1.45	.00	1.19**
Drive	13.18	3.49	12.76	3.24	.01	6.39**
Past focus	4.00	2.00	3.01	1.71	.02	3.78
Present focus	16.10	3.30	18.09	3.67	.03	15.28
Future focus	2.65	1.41	3.53	2.27	.03	1.45**
Swear words	0.32	1.07	0.54	0.69	.00	0.25

*$p < .0125$, **$p < .0038$ (significantly different from actual mean). LIWC: Linguistic Inquiry and Word Count.
Note: η_p^2 refers to the comparison between actual and cinematic speeches.

present in more than half of the speeches were environment reference, repeated words, in-speech instructions, game instructions, questioning, and metaphor. With the exception of anecdote and team past reference, all of the remaining characteristics were present in at least one-quarter of the cinematic speeches.

A set of chi-square tests of independence was performed to examine differences between actual and cinematic speeches on the verbal content characteristics. As displayed in Table 2, statistically significant differences between actual and cinematic speeches were found for opponent reference, team past reference, repeated words, in-speech instructions, game instructions, and metaphor. With the exception of metaphor, all of the characteristics for which significant differences were found were more prevalent in actual speeches than in cinematic speeches. A similar set of chi-square tests of independence was performed to examine differences between high school and intercollegiate pregame speeches on the verbal content characteristics. As shown in Table 2, a statistically significant difference between groups was found for only one verbal content characteristic, questioning, which was used more frequently in high school pregame speeches than in intercollegiate pregame speeches.

LIWC2015 variables

Means and SDs of LIWC2015 variables are presented in Table 3. In the one-way MANOVA comparing the actual

speeches with the cinematic speeches on the four LIWC2015 summary variables, a nonsignificant multivariate effect of speech type was obtained, Wilks' lambda = 0.97, $F(4, 113) = 0.81$, $p = .52$, partial eta-squared = .01. Similarly, in the MANOVA comparing the actual speeches with the cinematic speeches on the 13 LIWC2015 variables of potential relevance to pregame speeches, the multivariate effect of speech type was also not statistically significant, Wilks' lambda = 0.89, $F(13, 104) = 0.89$, $p = .46$, partial eta-squared = .11.

In the one-way MANOVA comparing the high school and intercollegiate pregame speeches on the LIWC2015 summary variables, a nonsignificant multivariate effect of competitive level was obtained, Wilks' lambda = 0.92, $F(4, 102) = 2.24$, $p = .07$, partial eta-squared = .08. In the one-way MANOVA comparing the high school and intercollegiate pregame speeches on the 13 LIWC2015 variables of potential relevance to pregame speeches, however, multivariate effect of competitive level was statistically significant, Wilks' lambda = 0.66, $F(13, 93) = 3.62$, $p < .001$, partial eta-squared = .34. As shown in Table 4, relative to high school pregame speeches, intercollegiate speeches had significantly higher we, positive emotion, negative emotion, present focus, and swear word scores.

A set of one-sample t-tests was conducted to compare the actual pregame football speeches and "natural speech" on the LIWC2015 variables. The natural speech sample included "diverse transcripts from multiple contexts, including people wearing audio recorders over days or weeks, strangers interacting in a waiting room, couples talking about problems, and open-air tape recordings of people in public spaces"[10] (p. 9). As shown in Table 3, the actual football pregame speeches had significantly higher analytical thinking, clout, we, you, they, negative emotion, drives, and future focus scores and significantly lower emotional tone, I, He/She, compare, and positive emotion scores than natural speech.

Table 4. Means and standard deviations of LIWC2015 variables for intercollegiate and high school pregame speeches.

	Intercollegiate		High school		
Variable	*M*	*SD*	*M*	*SD*	η_p^2
Analytical thinking	39.33	20.14	33.18	16.34	.03
Clout	93.69	11.60	94.54	12.16	.00
Authentic	55.49	22.19	55.55	21.47	.00
Emotional tone	62.84	33.75	75.90	22.09	.06
I	2.67	2.56	2.71	2.84	.00
We	4.83	2.78	3.62**	2.18	.06
You	5.24	2.23	6.17	2.69	.03
He/She	0.37	0.70	0.42	0.70	.00
They	1.05	1.30	1.23	1.25	.00
Compare	2.07	1.50	1.81	1.14	.01
Positive emotion	5.17	2.23	4.33**	1.58	.05
Negative emotion	2.59	2.12	1.16**	0.91	.18
Drive	14.08	3.64	12.72	3.34	.03
Past focus	4.05	2.30	3.98	1.85	.00
Present focus	14.91	3.24	16.72**	3.17	.07
Future focus	2.29	1.27	2.83	1.45	.03
Swear words	0.74	1.74	0.10**	0.17	.08

*$p < .0125$, **$p < .0038$ (significantly different from intercollegiate mean).
LIWC: Linguistic Inquiry and Word Count.
Note: η_p^2 refers to the comparison between intercollegiate and high school speeches.

Discussion

The main purposes of the current study were: (a) to describe the content, delivery, and environmental context of pregame speeches in American football and (b) to compare the speeches across different (i.e. high school and intercollegiate) athlete populations and with both ordinary speech and pregame speeches featured in motion pictures. Toward accomplishment of these aims, the present study was an in-depth analysis of the content, delivery, and environmental context of pregame speeches in American high school and intercollegiate football. American football pregame speeches were observed, coded, and analyzed through a descriptive coding system and LIWC2015 software. The speeches were compared with cinematic American football speeches and natural speech. From a purely descriptive standpoint, the results provide a clear portrait of the content and delivery of the speeches and the context in which they were delivered. The high school and intercollegiate speeches typically were made by team coaches, lasted approximately 2 min, and took place in a locker room setting with the speaker either positioned in front of or in the middle of the team while speaking. All of the high school and intercollegiate speeches included repeated words, and most also included questioning, issued game instructions, and made reference to the opponent, the team's past, and one or more aspects of the physical environment. The speeches contained confident, emotion-rich language. Thus, as anticipated, the content and delivery of the speeches provide support for the emphasis on informational and emotional aspects of pregame speeches in previous research[1,2,6–8] and are highly compatible with a situation in which coaches are addressing a team prior to a game against an opponent.

Comparisons of the American football pregame speeches across athlete populations and with both ordinary speech and cinematic pregame speeches revealed both similarities and differences. When comparing speeches delivered to high school football players to those delivered to intercollegiate football players, differences were found on 1 descriptive and verbal content variable (i.e. questioning), none of the LIWC2015 summary variables, and 5 of the 13 LIWC2015 nonsummary variables. High school pregame speeches included more present-oriented verbs (i.e. present focus) than intercollegiate pregame speeches. In contrast, intercollegiate football pregame speeches

contained more we, positive emotion, negative emotion, and swear words than high school pregame speeches. Swear words may have been more prevalent in intercollegiate pregame speeches because such language is often perceived as being more acceptable for older (adult) players. Swear words may increase player intensity and motivation, but they also may bring about negative emotions in the listener.[22] The greater presence of swear words in intercollegiate pregame speeches than in high school pregame speeches may have prompted the differences in negative emotion between the two levels of play, as swear words often have negative psychological associations. Thus, although the pregame speeches given to high school teams were substantially similar to those given to intercollegiate teams, it is possible that the relatively few variables on which differences were observed between the speeches delivered to the two groups reflect an attempt of speakers to tailor their messages to the age and/or developmental level of their respective audiences. This possibility warrants further investigation.

Comparisons between the actual speeches and natural speech revealed differences on 14 of the 17 LIWC2015 variables assessed. These results augment the more general descriptive findings and strongly suggest that the language used in speeches made prior to American football games constitute a distinct form of communication that appears to be tailored to the unique demands of the situation. Characterized by use of confident, analytical, motivational, emotionally tinged language that is collectively oriented and present and future focused, the American football pregame speeches examined in the current study seem to clearly reflect the common foci of the speeches on performing well in an imminent competitive event.

Relative to the cinematic speeches, the actual speeches contained more references to several football-related and team-related aspects (i.e. team past and opponent references, game instructions) and less use of metaphor. This finding is attributable to the divergent foci of the two kinds of speeches, with actual speeches focused primarily on the game in question and the cinematic speeches focused mainly on advancing the plot. The substantial similarity between the actual and cinematic speeches on the LIWC2015 variables (as evidenced by the low partial eta-squared values and lack of statistically significant differences) attests to the realism of the cinematic speeches (and/or a possible reciprocal influence between actual and cinematic speeches). Overall, results of the analyses comparing actual and cinematic pregame speeches suggest the existence of a popular narrative of what a pregame speech is and how it should be delivered.

Several limitations should be considered when interpreting the current findings. First, the presence of a recording device may have prompted the coaches and other speakers to alter the content of their pregame addresses. Given the current ubiquity of electronic devices with recording capability, however, it is reasonable for those delivering pregame speeches to anticipate the possibility of their remarks being recorded. Second, the sample of pregame speeches was restricted to those that had been uploaded to YouTube. It is not known how representative the pregame speeches appearing on YouTube are of high school and intercollegiate American football speeches. It should be noted, however, that the diverse sample of speeches that were analyzed represented the population of speeches available on the platform on the date of data collection. Third, also with respect to generalizability, it is unclear how well the current findings generalize to pregame speeches at other competitive levels of American football (e.g. professional, youth), in other sports, and with other genders. The generalizability of the present results should be examined in subsequent investigations. Fourth, although not a focus of the current investigation, the extent to which the pregame speeches had the motivational and performance effects that they were intended to produce is not known. Further experimental inquiry is needed to investigate this issue. Fifth, the focus of the current research was primarily descriptive and the findings were by design largely independent of theoretical considerations. As the characteristics of pregame speeches are elucidated, however, theoretical perspectives on emotions, group dynamics, leadership, learning, motivation, persuasion, and speech communication can be used to enrich research by providing testable hypotheses.

Beyond the need to address the limitations of the current study, the descriptive analysis of American football pregame speeches given in naturalistic settings has important implications for future research. The descriptive and psycholinguistic analyses conducted in the current investigation can be used not only in experimental studies to develop stimulus materials (i.e. fictional pregame speeches) that differ along dimensions found to correlate with motivational and/or performance variables of interest. Systematic fine-grained evaluations of athletes' preferences for various descriptive and psycholinguistic aspects of pregame speeches could also be conducted to enable investigators to build upon seminal research on the topic[1,2,5–9,23] and, ultimately, facilitate the development of more effective pregame speeches.

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